

Death by polonium-210: lessons learned from the murder of former Soviet spy Alexander Litvinenko.

The medical response to radiation--whether the result of radiological warfare, terrorist deployment of improvised radiation dispersal weapons, political assassination, occupational or industrial accidents or the medically radiated patient remains one of the least taught among all disciplines within medical education. In the aftermath of 9/11 among medical vulnerabilities to toxicant threats, of all the categories of weapons of mass destruction (WMD)--whether using the CBRNE (chemical, biological, radiological, nuclear, explosive) or NBC (nuclear, biological, chemical) acronym--radiation is the least taught in professional schools, responder cultures or civil preparedness organizations. To date, few health care professionals (HCP) possess the fundamental knowledge or skills to identify and diagnose, let alone treat a radiation victim; this vulnerability made even more obvious in the aftermath of the high profile assassination of former Russian agent Alexander Litvinenko. He was poisoned with Polonium210. Radioactive substances are ubiquitous with radiation sources being in or transported through virtually every region nationwide. It is essential to increase preparedness among community and rural health care facilities as well as urban and university hospitals. Managing radiation injuries effectively requires access to specialized equipment and expertise. Radiation sickness is progressive and may require acute, critical and long-term care throughout the course of illness. Regardless of the source, preparedness rests upon acknowledging a threat exists and dedicating the resources to address the risks including the enhancement of training and equipment. Mass or individual exposures to radiation present unique challenges to the entire response continuum from law enforcement, first responders and emergency medical care. Increased education about and practice in responding to radiological threats is essential to enhance preparedness.